

Harkirat Singh

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Professional Summary

Software Engineer specializing in Generative AI and backend architecture, with a proven track record of building production-ready AI applications and scalable backend systems. Expert in Python, FastAPI, and LangChain, with deep practical experience architecting multi-agent orchestration frameworks, RAG pipelines, and automated semantic search workflows. Adept at applying modular clean architecture principles, managing vector databases, and engineering reliable REST APIs designed for modern production AI environments.

Technical Skills

Languages	Python, SQL (Structured Query Language)
Backend	FastAPI, REST APIs, Asynchronous Programming (AsyncIO), SQLAlchemy, Pydantic, JWT Authentication
GenAI	LangChain, LCEL (LangChain Expression Language), Prompt Engineering, AI Agents, Multi-Agent Systems, RAG
LLM APIs	OpenAI API, Google Gemini API, NVIDIA API, Hugging Face Transformers, AI Automation
Vector DBs	ChromaDB, Pinecone, FAISS, Semantic Search, Embeddings
Databases	PostgreSQL, SQLite, MySQL
Cloud	AWS (EC2, S3, RDS), Production AI Systems
Tools	Docker, Git, Clean Architecture, Pandas

Internship Experience

Backend Developer Intern <i>AI-Powered Website SEO Audit Platform</i>	<i>Jul 2026 – Present</i> Pune, India
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- Engineered a high-throughput backend using Python and FastAPI, establishing secure REST APIs, scalable services, and asynchronous polling pipelines for end-to-end, multi-page website analysis.
- Architected an asynchronous website crawler leveraging `asyncio` and BeautifulSoup, accelerating raw data ingestion while enforcing dynamic deduplication and automated `robots.txt` protocol compliance.
- Implemented semantic analysis engines using PostgreSQL and SQLite to execute complex on-page metadata extractions, structured data evaluations, and validation loops for structural heading integrity.
- Integrated OpenAI and Google Gemini APIs via an isolated LLM prompts layer, enabling runtime generation of context-aware SEO optimization scores, priority issue lists, and tailored copy recommendations.
- Designed a modular, decoupled architecture (routers, services, vector-agents, repositories) featuring automated PDF and Excel rendering modules to cleanly isolate processing workloads from report distribution.

Projects

Multi-Agent AI Platform with Fallback LLM Routing <i>LangChain Multi-Agent Orchestration Framework</i>	<i>Jan 2026 – Mar 2026</i> Personal Project
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- Built a multi-turn LCEL orchestration platform that dynamically routes user queries across 3 custom-built tool layers (Web Search, Calculator, and SQLite DB) using persistent session state, successfully converting a 200+ test suite with a 97% task-completion rate.
- Optimized operational uptime by configuring Google Gemini as the core inference system backed by instant fallback routing to NVIDIA-hosted open-source models, mitigating API rate-limiting and lowering compute expenditures.
- Developed custom tool-calling and function-calling schemas wrapped inside exponential-backoff retry layers, decreasing unhandled system exceptions from 12% down to under 1% across 1,400+ continuous integration invocations.
- Conducted systematic RAG evaluation comparing multi-step reasoning accuracy metrics, demonstrating a verifiable performance gain from 74% to 91% upon deploying persistent short-term context windows.

Certifications

Fundamentals of Machine Learning and Artificial Intelligence – AWS Training & Certification	July 2026
Foundations of Prompt Engineering – AWS Training & Certification	July 2026
Amazon Connect Routing Fundamentals – AWS Training & Certification	July 2026

Education

Bachelor of Engineering in Computer Engineering <i>Sinhgad Institute of Technology</i>	<i>2023 – 2027</i> Pune, India
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